KARISHMA SAPKOTA

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RESEARCH SUMMARY

Interdisciplinary researcher specializing in clinical informatics and human-centered AI, integrating a professional nursing foundation with advanced training in data science to design computational systems that turn heterogeneous healthcare data into clinically actionable insights. I am particularly interested in leveraging multimodal data fusion, linking unstructured clinical narratives (NLP) with structured physiological and operational data to build explainable, scalable intelligent systems that support clinician workload, documentation efficiency, and real-time clinical decision support. My long-term goal is to develop transparent AI frameworks that enhance disease surveillance, resource allocation, and frontline usability, strengthening health system resilience in both U.S. and global healthcare settings.

RESEARCH INTERESTS

- Biomedical and Health Informatics
- Machine Learning and AI for Bioinformatics
- Natural Language Processing (NLP)
- Intelligent Systems and Clinical Decision Support
- Human-AI Interaction and Explainable AI
- Multimodal Health Data Integration and Visualization

EDUCATION

Master of Science in Data Science 08/2022-05/2024

Mercyhurst University, Erie, PA, USA

- **Mathematical Foundations:** Calculus, Linear Algebra, Probability and Statistics.
- **Computational Research:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Algorithm Development, Big Data Analytics.
- **Tools:** Python (PyTorch, TensorFlow, Scikit-learn), SQL, PySpark.
- **Graduate Assistantship:** Selected to co-develop NLP curriculum and facilitate student instruction, specializing in transformer-based architecture.

Bachelor of Science in Nursing 12/2012-12/2016

Purbanchal University, Nepal

- **Honors:** Ranked 3rd in Academic Standing (Entire cohort).
- **Relevant Coursework:** Health Informatics, Research Methodology, Biostatistics, Clinical Decision Support Systems (CDSS), Ethics in Healthcare Research.

RESEARCH & HEALTH SYSTEMS ANALYTICS EXPERIENCE

Operational Research Contributor-Neglected Tropical Diseases (NTDs) 03/2020-05/2022

Crown Agents Ltd./ASCEND (Accelerating the Sustainability Control & Elimination of Neglected Tropical Diseases) Program, Nepal (Remote)

- **Spatio-temporal analytics:** Analyzed provincial NTD datasets to identify high-risk geographic clusters, temporal transmission patterns, and program performance gaps, informing targeted intervention strategies.
- **Clinical data profiling:** Conducted retrospective chart reviews and municipal disease profiling to evaluate service delivery quality for Lymphatic Filariasis and Visceral Leishmaniasis.
- **Decision-support reporting:** Synthesized complex epidemiological and operational data into structured analytical summaries used by provincial health authorities for strategic planning and resource allocation.
- **Data Workflows:** Designed data workflows improving surveillance accuracy and reporting consistency across multiple districts.
- **Health systems informatics:** Contributed to the design of data workflows that improved surveillance accuracy and reporting consistency across multiple districts.

Consultant-Clinical Outcomes Research (Ultrasound vs. No Ultrasound Study) 04/2019-06/2019

Ipas Nepal

- **Data abstraction & QA:** Retrieved, abstracted, and validated multi-facility clinical outcome data, implementing rigorous quality-assurance protocols to improve dataset reliability for comparative analysis.

- **Ethics & compliance:** Ensured strict adherence to research ethics, confidentiality standards, and clinical documentation protocols across hospitals and health posts.
- **Translational insights:** Supported preliminary statistical analyses that informed evidence-based recommendations for improving second-trimester abortion care in low-resource settings.

Research Assistant (Mixed-Method Health Systems Study) 01/2019-04/2019
Oxford Policy Management (OPM), Nepal

- **Informatics System Synthesis:** Integrated primary field data with secondary Health Management Information System (HMIS) datasets to develop comprehensive health profiles for municipal assessment.
- **Qualitative Analytics:** Conducted verbatim transcription and thematic coding of in-depth interviews with municipal stakeholders to identify operational bottlenecks and governance gaps.
- **Data Integrity & Verification:** Spearheaded quantitative data verification and cleaning for Basic Health Service (BHS) utilization indicators, ensuring metric accuracy for national policy discussions.

Malaria Elimination Support Program-Data & Program Support Officer 11/2017-03/2018
InfoAids Nepal Funded by Save the Children & Supported by EDCD (Epidemiology & Disease Control Department) Nepal

- **Surveillance Workflow Design:** Developed data-driven tools and workflows to strengthen national malaria surveillance and case investigation across high-risk districts.
- **Operational Standardization:** Co-authored the National Malaria Elimination Operational Manual, standardizing nationwide data reporting and intervention procedures in collaboration with government stakeholders.
- **Monitoring & Evaluation (M&E):** Designed and implemented monitoring systems to track intervention coverage and supply chain readiness, utilizing field data to identify epidemiological hotspots.

INDEPENDENT RESEARCH

Comparative Affective Computing in Digital Mental Health (2025)
Project Focus: Computational Linguistics & Mental Health Informatics

- Evaluated the performance of transformer-based vs. rule-based models in detecting psychological distress signals (anger, fear, sadness) within a dataset of 37,000+ unstructured narratives.
- Developed a Python-based pipeline comparing DistilRoBERTa with VADER and TextBlob, utilizing Ekman's taxonomy to map complex emotional states.
- Conducted a rigorous performance trade-off analysis between model accuracy and computational latency, identifying optimal architectures for mental health surveillance.
- Authored a comprehensive technical report detailing the findings, used as a primary evidence of research methodology and data engineering proficiency for this application.
- ***Note:*** *This project serves as the primary writing sample for this application.*

Undergraduate Thesis – Department of Nursing
Purbanchal University 5/2016-12/2016
Title: Research on Menstrual Hygiene Knowledge and Practice in Women of Reproductive Age, Jamtoki Morang, Nepal
Role: Lead Independent Researcher (Supervisor: Prisha K.C., M.S.)

- Led a cross-sectional study (n=200) on menstrual hygiene knowledge and practices among women aged 15–49.
- Designed and validated a bilingual (Nepali & English) questionnaire and managed full research pipeline, including ethics and field data collection.
- Performed quantitative analysis in SPSS, identifying links between socioeconomic/education status and hygiene practices, and presented findings to institutional leadership.

TECHNICAL & DATA SCIENCE PROJECTS

Sentiment Analysis on Yelp, IMDB, and Amazon Review

- Built an NLP (Natural Language Processing) based sentiment classifier using Yelp, IMDB, and Amazon datasets with NLTK, SpaCy, TF-IDF, and word embeddings.
- Trained and evaluated SVM, Naive Bayes, Logistic Regression models achieving an 86% accuracy in sentiment classification.

Clinical Predictive Modeling (Heart Disease)

- Developed a predictive model for early detection of heart disease using datasets (Kaggle).
- Processed and analyzed physiological and demographic features and applied a Decision Tree Classifier to model the relationship between features such as age, sex, blood pressure, cholesterol levels, and other critical medical metrics.
- Evaluated model performance using confusion matrix, precision, recall, and F1-score, ensuring accuracy and minimizing biases in predicting heart disease presence.

Imbalanced Data Classification (Fraud Detection)

- Developed a logistic regression model to accurately differentiate between fraudulent and non-fraudulent credit card transactions in a highly imbalanced dataset available at Kaggle.
- Evaluated model effectiveness using the Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC), achieving a high accuracy of 95% in distinguishing fraudulent transactions.

PROFESSIONAL EXPERIENCE

Business Analyst 07/2025- Present

YuVi Systems, Remote (Consultant)

- **Requirements Engineering:** Collaborate with cross-functional stakeholders to translate complex organizational needs into technical analytical requirements and standardized KPIs.
- **Exploratory Data Analysis (EDA):** Execute Exploratory Data Analysis (EDA) on high-dimensional datasets to identify latent patterns in operational efficiency.
- **Knowledge Translation:** Synthesize technical findings into structured analytical summaries for diverse audiences, bridging the communication gap between technical developers and non-technical leadership.

Business Analyst (Clinical Informatics) 09/2024- 06/2025

UPMC Hamot-Pharmacy Department, Erie, PA

- **Advanced Clinical Decision Support (CDS):** Architected and deployed a suite of enterprise-level Power BI dashboards for real-time monitoring of pharmacy workflows. This included tracking pharmacist verification cycles, order volume spikes, and staff workload distribution to optimize hospital-wide medication delivery.
- **End-to-End Data Engineering & Integration:** Engineered robust data pipelines using SQL, Power Query, and DAX to integrate fragmented data from pharmacy management systems, clinical databases, and operational logs. Ensured high-fidelity data harmonization to support executive-level strategic decisions.
- **Safety & Quality Assurance (Technician Workflow):** Designed a centralized medication error monitoring system by integrating Microsoft Forms with Power BI. This created a real-time feedback loop specifically for tracking Pharmacy Technician workflows, allowing leadership to identify risks in drug storage, expiration management, and drug-mixing protocols for immediate corrective action.
- **Medication Safety Surveillance:** Developed automated surveillance dashboards to track pharmacist interventions and "near-miss" error prevention activities, strengthening monthly safety reviews and supporting continuous quality improvement (CQI).
- **Operational KPI Analytics:** Conducted deep-dive analysis into critical performance indicators, including prescription turnaround time, medication stock-out rates, and staff productivity metrics. Translated these findings into actionable workflow recommendations that reduced bottlenecks in the medication distribution chain.
- **Predictive Modeling & Inventory Optimization:** Applied predictive analytics and time-series forecasting to optimize pharmaceutical procurement. This ensured 100% availability of high-priority medications while maintaining strict regulatory compliance and minimizing waste through improved inventory turnover.
- **Process Automation & Efficiency:** Designed and implemented automated workflows using Power Automate, reducing manual report generation time by 25–30%. This shift significantly reduced information latency, providing pharmacy managers with near-real-time operational visibility.
- **Information Governance:** Established and maintained a centralized SharePoint Analytics Hub, standardizing access to dashboards, data dictionaries, and technical documentation. This served as a "single source of truth" for pharmacy leadership and multidisciplinary clinical teams.
- Maintained strict **HIPAA compliance** during the engineering of clinical data pipelines, ensuring the de-identification of Protected Health Information (PHI) within large-scale hospital databases.

Provincial Neglected Tropical Disease Coordinator 03/2020-05/2022

Crown Agents, United Kingdom| Remote

- **Health Information Systems & Data Integration:** Designed and maintained provincial data management workflows to harmonize fragmented municipal datasets for Neglected Tropical Diseases (NTDs). Analyzed surveillance data using SQL and SPSS to identify high-risk geographic clusters for Lymphatic Filariasis and Visceral Leishmaniasis.
- **Epidemiological Surveillance & KPI Analytics:** Led provincial monitoring of key performance indicators, including intervention coverage, treatment completion, and diagnostic timeliness. Developed analytical dashboards that translated field data into actionable insights for health authorities.
- **Operational Research Contributions:** Conducted retrospective chart reviews and municipal disease profiling to assess service delivery quality. Findings supported evidence-based planning for Nepal's national strategies on Kala-azar and Lymphatic Filariasis elimination.
- **Clinical Supply Chain Monitoring:** Coordinated data-driven logistics for diagnostic kits and essential medications by tracking inventory levels, distribution timelines, and district-level demand patterns across high-burden areas.
- **Field Supervision & Data Quality Assurance:** Oversaw district surveillance teams during school-based LF mapping activities. Implemented protocols for ethical sample collection, secure data handling, and timely clinical follow-up for positive cases.
- **Capacity Building & Data Governance:** Trained provincial and municipal health staff in Monitoring & Evaluation (M&E), data quality assurance, and standardized reporting workflows, improving consistency and reliability of surveillance data.
- **Strategic Planning & Coordination:** Prepared provincial implementation plans, budgets, and reporting packages, aligning international donor requirements (Crown Agents/FCDO) with local government health system capacities.

Nursing Officer 07/2017-02/2020

Department of Health, Nepal

- **Clinical Data Stewardship & Analytics:** Developed and maintained a longitudinal SPSS database to track reproductive health outcomes; conducted quantitative analysis of maternal health indicators to identify service gaps and inform evidence-based program planning.
- **Health Information Systems (HIS) Management:** Designed and standardized medical record-keeping workflows across facilities to improve data integrity and completeness for longitudinal monitoring of high-risk conditions.
- **HIV/AIDS Program Implementation:** Managed patient registries for Voluntary Testing and Counseling (VCT) and Prevention of Parent-to-Child Transmission (PPCT) services, monitoring treatment adherence and program performance across multiple hospitals.
- **Policy & Service Standardization:** Contributed clinical insights to national working groups for Safer Motherhood (SM) and Skilled Birth Attendant (SBA) policy updates, aligning frontline clinical data with national delivery standards.
- **Training & Capacity Building:** Collaborated on national-level training modules for Emergency Obstetric Care (BEmONC); served as Lead Trainer for clinical protocols and standardized documentation workflows across multiple districts.
- **Community Health Outreach:** Coordinated mobile health camps and community-based interventions, overseeing field data collection to assess health service needs among underserved populations.

TEACHING EXPERIENCE

Guest Lecturer, Database and SQL Web Security (MS Cybersecurity) 08/2024

Mercyhurst University, Erie, Pennsylvania

- Delivered a hands-on session on SQL querying, including SELECT, JOIN operations, data filtering, and query optimization.
- Explained why foundational SQL skills are critical for understanding later topics such as SQL injection and web security.
- Connected foundational SQL skills with cybersecurity concepts to prepare students for the subsequent lecture on web security and defensive programming.

Graduate Assistant 01/2023-05/2024

Computer and Information Science Department, Mercyhurst University, Erie, PA, USA

- Co-developed graduate-level NLP course materials by synthesizing academic resources and preparing standardized code examples, lab notebooks, and instructional pipelines now used by faculty.
- Built modular NLP pipelines for sentiment classification, text summarization, and NER using rule-based and transformer-based architectures.
- Maintained a shared repository of sample code, algorithmic walkthroughs, and annotated notebooks to support consistent instruction across course sections.
- Assisted students with NLP concepts, debugging, and translating theoretical algorithms into practical Python implementations.

Guest Lecturer, Nursing (3rd year Students) 02/2016-03/2016

Purbanchal University, Nepal

- Taught dermatological diseases, including etiology, prevention, and clinical management.
- Designed and conducted exams to assess students' understanding and applied knowledge.
- Enhanced students' comprehension of clinical conditions and patient care strategies.

Undergraduate Curriculum Development – Med-Surg & Clinical Rotations (1st–3rd Year Students) |

Purbanchal University, Nepal

- Developed a structured curriculum for medical-surgical nursing and clinical rotations for 1st–3rd year nursing students.
- Designed learning objectives, clinical skill checklists, and evaluation rubrics.

Community Health Educator 08/2015

- Conducted educational sessions in local communities on maintaining hygiene and preventing mosquito-borne diseases.
- Promoted awareness and behavior change through interactive demonstrations and discussions.

TECHNICAL SKILLS

- **Programming & Databases:** Python, R, JavaScript, SQL Server, PostgreSQL, MongoDB (NoSQL), data modeling
- **Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost; Random Forest, SVM, Decision Trees, Linear & Logistic Regression; CNN, RNN; Computer Vision, Explainable AI
- **Natural Language Processing (NLP):** NLTK, SpaCy, TextBlob, BERT, Word2vec, Gensim, GloVe, LangChain
- **Health Informatics & Research:** Hypothesis generation, survey design & validation; mixed-methods (key informant interviews, focus groups, structured/unstructured questionnaires, observational studies); health systems research; program monitoring & evaluation; data-driven policy analysis
- **Statistical Modeling:** SPSS, SAS; hypothesis testing, exploratory data analysis (EDA), PCA, correlation/covariance analysis, time series analysis, regression (linear, logistic, multinomial), discriminant analysis
- **Data Engineering & Processing:** Web scraping (Scrapy, BeautifulSoup, Requests); Pandas, NumPy, SciPy; ETL processes; data cleaning and manipulation; Excel
- **Data Visualization:** Power BI, Tableau, Plotly, Matplotlib, Seaborn, heatmaps
- **Data Governance & Ethics:** HIPAA Compliance, Research Ethics

CERTIFICATIONS & TRAININGS

Technical & Data Science Certifications

- **IBM:** Big Data, Machine Learning with Python, Python for Data Science, SQL & Relational Databases
- **LinkedIn Learning:** Advanced AI–NLP Techniques for Clinical Datasets, Generative AI with GPT-4 & LangChain
- Power BI Data Modeling with DAX, Advanced Microsoft Power BI, Power BI & BIM Analytics
- **HackerRank:** SQL Skill Certification
- **Techcanvass:** Agile Scrum Essentials

Research & Health Systems Trainings

- **Nepal Health Research Council:** Data Management & Analysis, Advanced Health Systems Research, Qualitative & Quantitative Research Methods
- Research Ethics (FHI 360)
- Leadership & Systems Thinking (Ministry of Health, Nepal)
- **M-TOT:** LF Morbidity Management, Mapping & Disability Prevention
- **M-TOT:** Malaria Elimination & Surveillance
- ASCEND Organizational Capacity Building
- Basic Health Service Delivery Systems at Sub-national Level

AWARDS AND HONORS

Academic Excellence Award: Ranked 3rd in Academic Standing (Top 3 of BSN cohort), Purbanchal University.

VOLUNTEER

- **Public Health Development and Research Center, Nepal:** Conducted reproductive health policy analysis and trained small groups of 10–12 youth advocates at multiple locations, selected based on their ability to actively advocate for health initiatives in their communities.
- **Gunjaman Singh Trust Hospital, Nepal:** Delivered patient care for underserved populations and managed clinical documentation in compliance with privacy and confidentiality standards.